EXPERIMENT 9: GIT AND GITHUB PULL REQUEST WORKFLOW

Aim

To demonstrate a Git and GitHub collaboration workflow by creating a repository, working on a feature branch, committing and pushing changes, creating a pull request, and merging the changes into the main branch.

Procedure

1. Create a public GitHub repository named CSA1023-Test-Repo with a README.md file.

2. Clone the repository to the local Software Engineering lab folder.

3. Check the repository status and create a branch named feature-update-readme.

4. Update README.md with information about the Software Engineering Lab and contribution workflow.

5. Configure the Git user name and email, then stage and commit the README.md changes.

6. Push the feature-update-readme branch to the remote GitHub repository.

7. Create a pull request with main as the base branch and feature-update-readme as the compare branch.

8. Review the changes and verify that GitHub reports no conflicts with the base branch.

9. Merge the pull request and confirm that it is successfully merged and closed.

Output

The screenshots below document repository creation, branch creation, the README update, the push operation, pull request creation, and the successful merge.

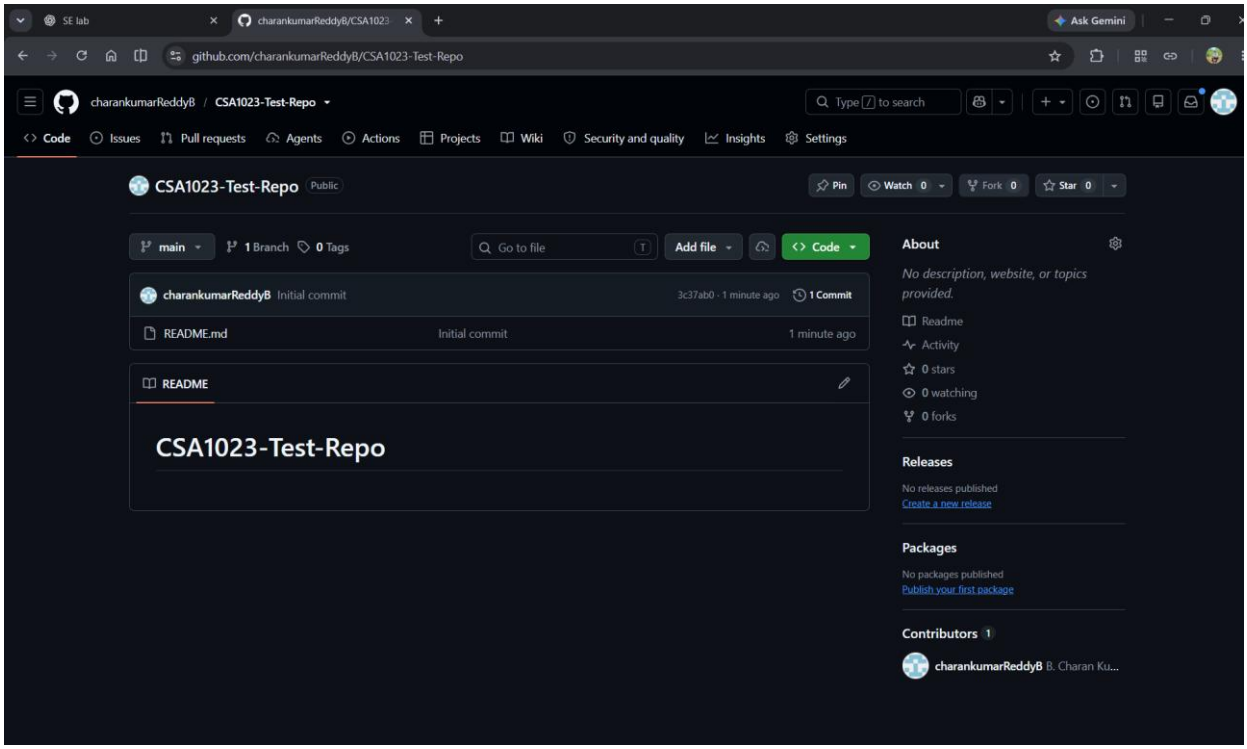


Figure 1: Created the CSA1023-Test-Repo GitHub repository.

```
PS C:\Charan\Simats\Software Engineering\lab> cd CSA1023-Test-Repo
PS C:\Charan\Simats\Software Engineering\lab\CSA1023-Test-Repo> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS C:\Charan\Simats\Software Engineering\lab\CSA1023-Test-Repo> git checkout -b feature-update-readme
Switched to a new branch 'feature-update-readme'
PS C:\Charan\Simats\Software Engineering\lab\CSA1023-Test-Repo> git branch
* feature-update-readme
  main
```

Figure 2: Created the feature-update-readme branch in the local repository.

```
PS C:\Charan\Simats\Software Engineering\lab\CSA1023-Test-Repo> git config --global user.name "CharanKumarReddyB"
PS C:\Charan\Simats\Software Engineering\lab\CSA1023-Test-Repo> git config --global user.email "charankumarreddybantr
othula@gmail.com"
PS C:\Charan\Simats\Software Engineering\lab\CSA1023-Test-Repo> git add README.md
>> git commit -m "Update README with contribution workflow"
>> git push -u origin feature-update-readme
[feature-update-readme 85178d1] Update README with contribution workflow
 1 file changed, 8 insertions(+), 1 deletion(-)
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 453 bytes | 453.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/charankumarReddyB/CSA1023-Test-Repo.git
   3c37ab0..85178d1  feature-update-readme -> feature-update-readme
branch 'feature-update-readme' set up to track 'origin/feature-update-readme'.
PS C:\Charan\Simats\Software Engineering\lab\CSA1023-Test-Repo>
```

Figure 3: Committed the README update and pushed the feature branch to GitHub.

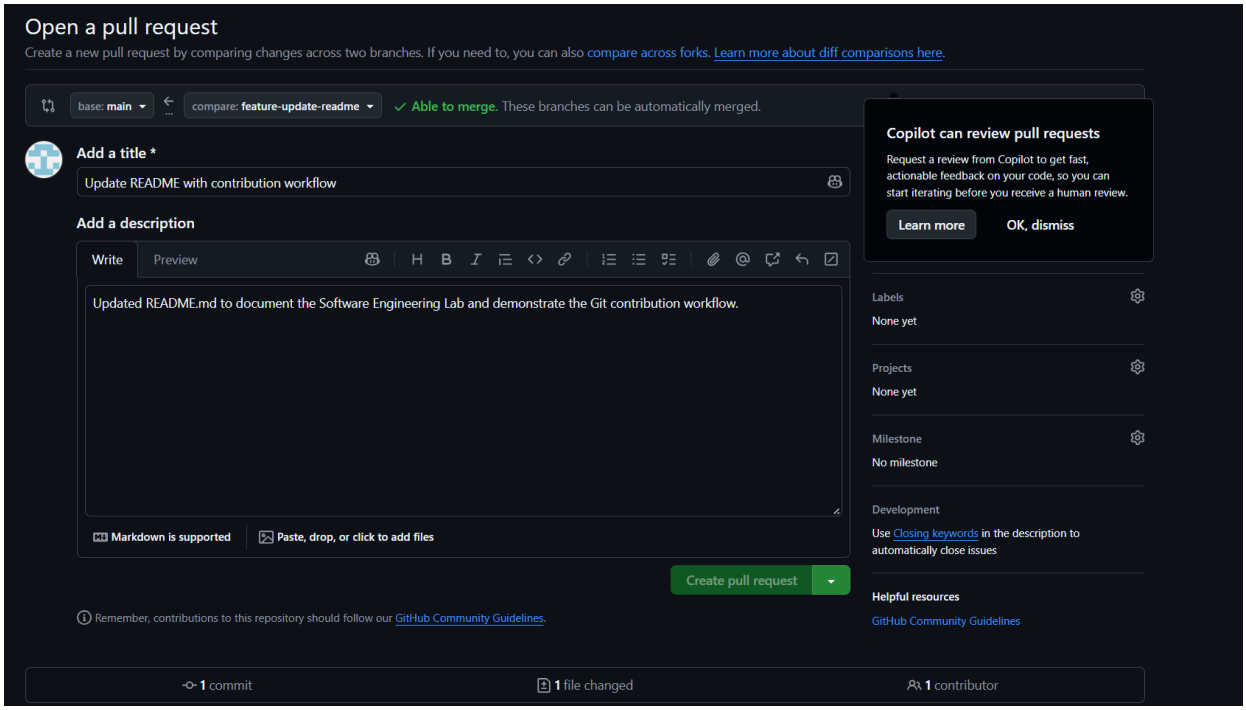


Figure 4: Compared main with feature-update-readme and prepared the pull request.

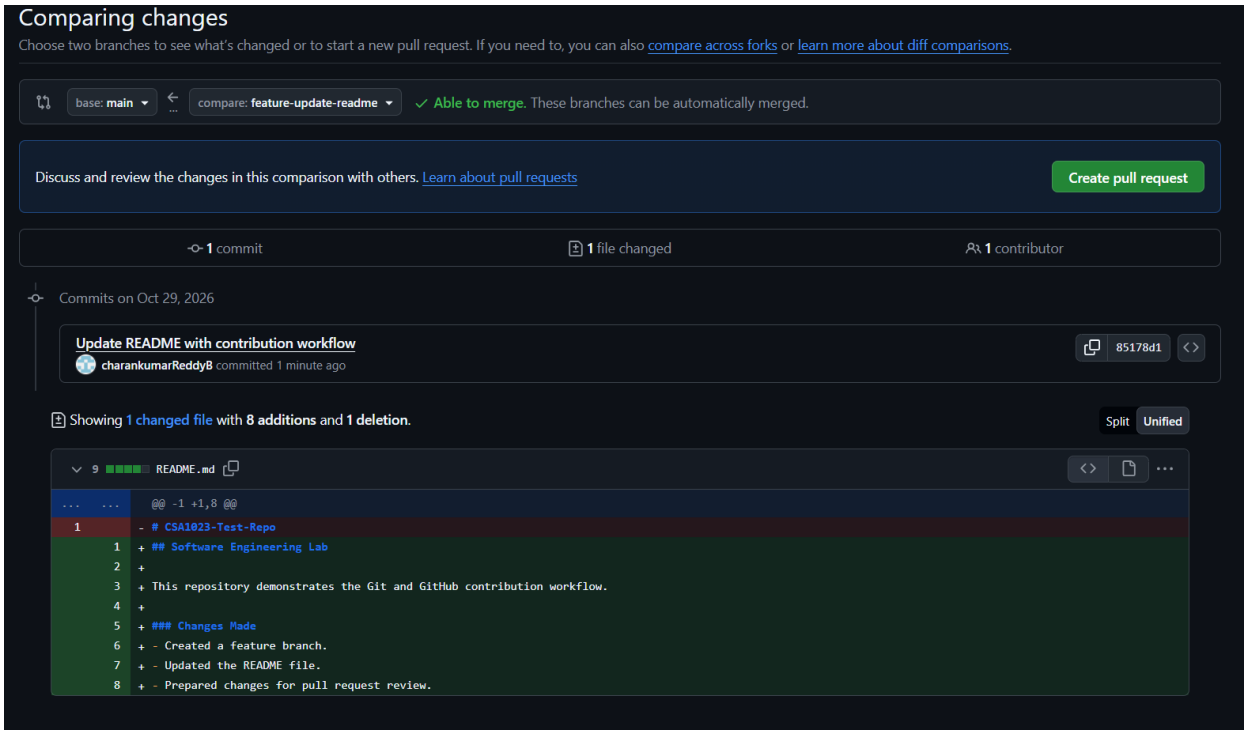


Figure 5: Pull request creation form and README changes.

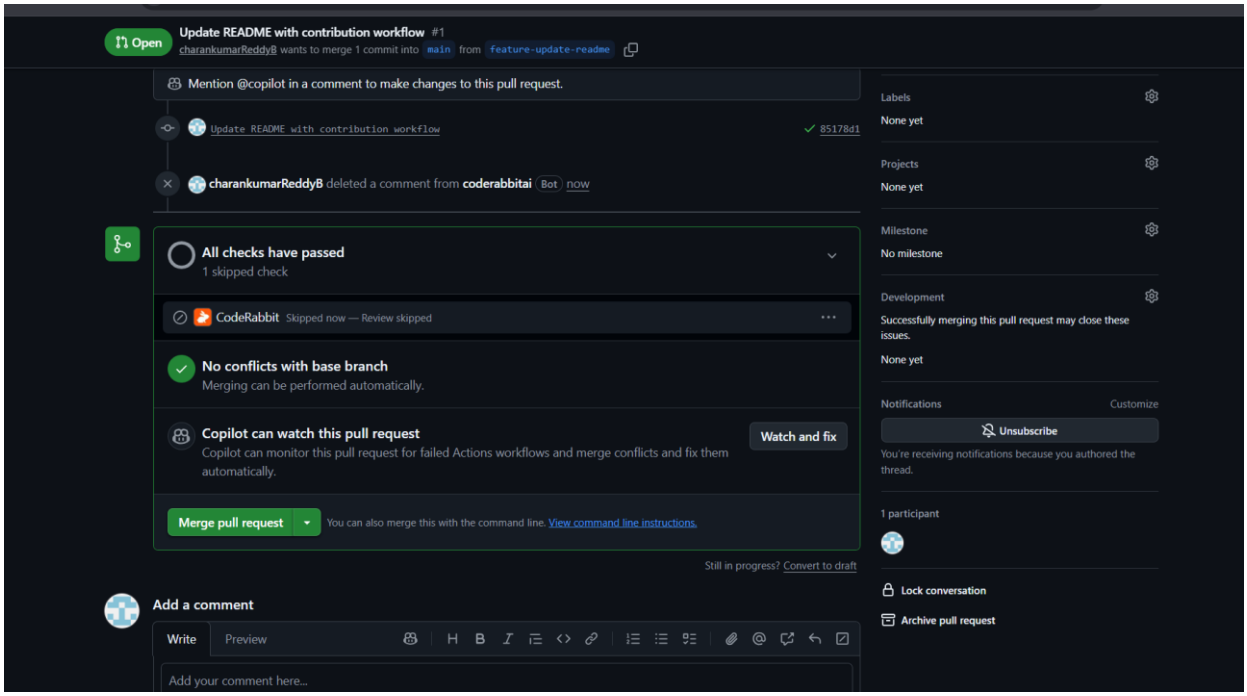


Figure 6: Pull request opened; checks passed and no merge conflicts were found.

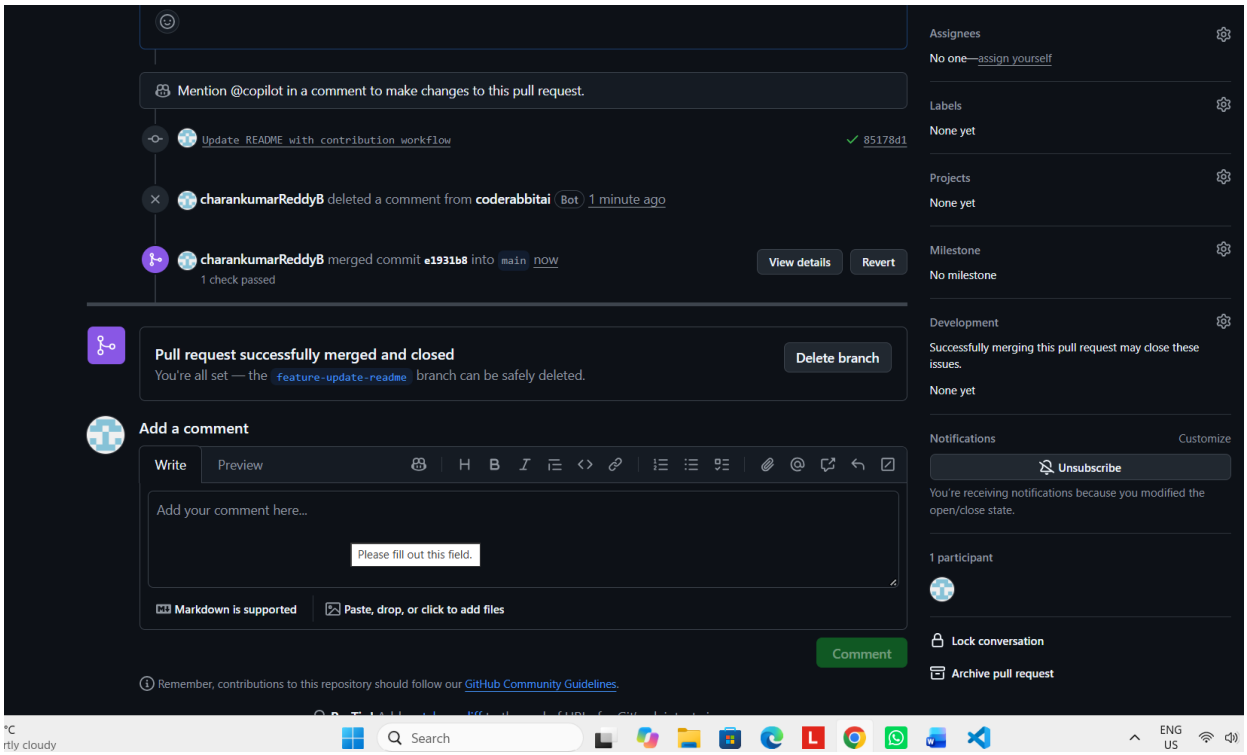


Figure 7: Pull request successfully merged and closed.

Result

Thus, the Git and GitHub branch-and-pull-request workflow was successfully demonstrated. The README changes were committed and pushed to the feature branch, submitted through a pull request, and merged into the main branch.